Initiation document for implementation

MaintControl

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0. Administration

0.1 Introduction - Existing situation (a problem or a gap definition)

The field of routine maintenance is a field that occupies a significant share of the organization's expenses.

`Also, routine maintenance is the most efficient way to ensure the safety of use and the integrity of the installed equipment.
For proper maintenance, a number of conditions must be met:

1. Skilled manpower to perform the maintenance.

2. The employee's arrival at the maintenance location at the scheduled time.

3. Perform routine maintenance in accordance with procedures established by the organization / manufacturer.

4. Effective flow of information about changes and / or additions to established procedures.

Today, in the field of maintenance there isn a problem of supervision and control over the maintenance that is carried out.
That is, there is a lack of:
On the one hand, in the platform that guides the maintenance man in the field, what should a maintenance person perform, in what order of operations and whether there have been Schanges and / or additions in the maintenance process.

On the other hand, the platform which provides a real-time measure of the control room and control over the maintenance performed.

0.1 Rationale for application development

By developing the application, gaps of:

1. Confession of routine maintenance, which takes place at the required time and extent.

2. Quality of routine maintenance.

3. Efficient flow of information related to field maintenance.

4. Collection of data for the scope of routine maintenance.

0.2 Concepts and Definitions

Location based application that will help maintenance service company to complete and optimize their routine work – using various parameters.

The application will help workers to do their job more effective due to knowledge database , work sequence and work documentation.

1. System description

1.1 The Character of an application

- An application is an application for collecting information from a field maintenance person, for the purpose of monitoring the level of routine maintenance.
- An identification component will be developed for the users.
- The application will have access to location services and the camera of the mobile device
- The application developed by the Mobile app, for the Android operating system
- The system will have a management interface (for "manager" and administrator users) adapted to the windows 10 operating system.
- The application will be developed as a Multilanguage application. The default language will be set by the administrator

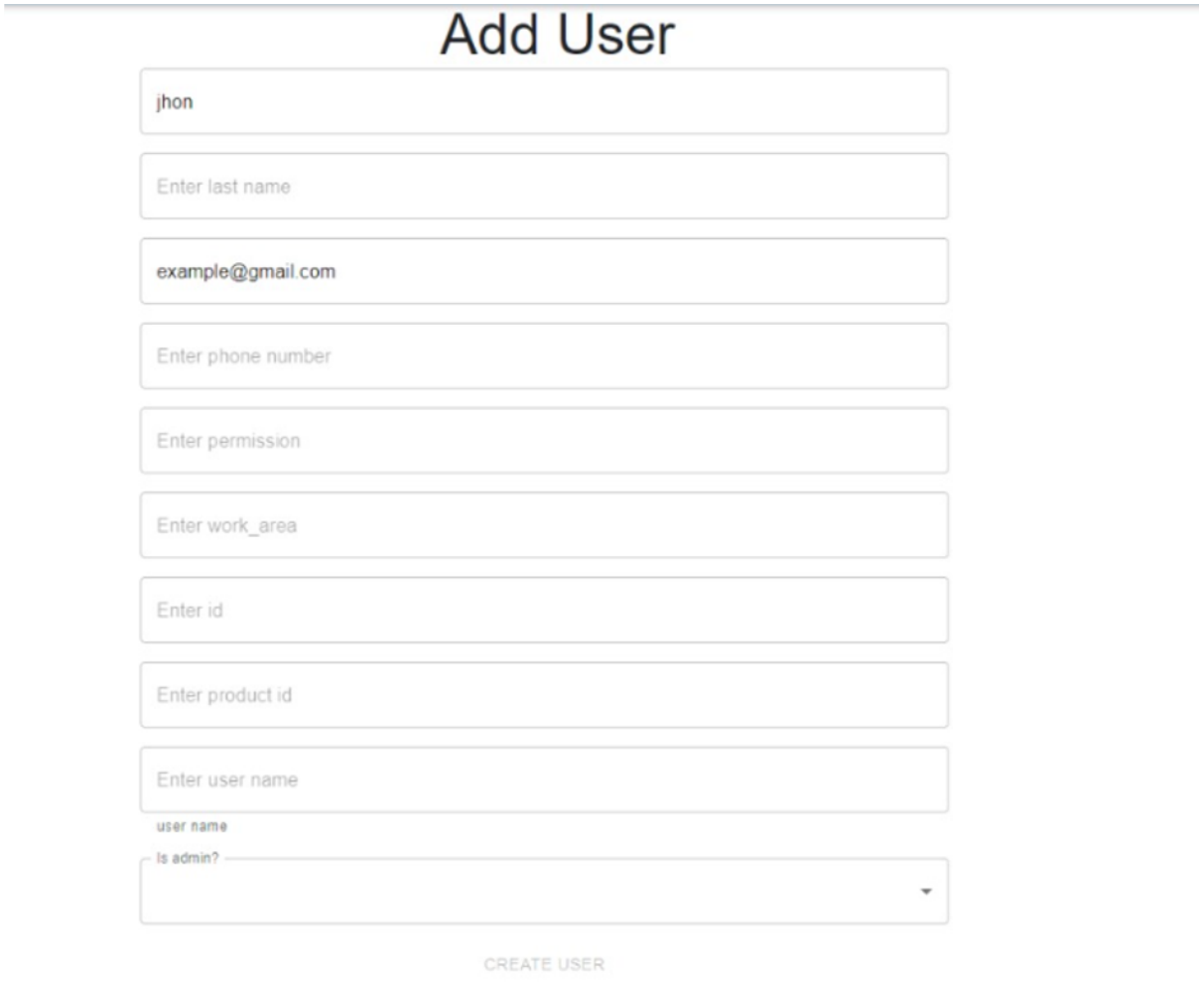
1.2 System users

"Maintenance man" user - a user who registers for the system, receives a list of maintenance tasks and indicates the execution of the tasks. The user will be able to enter comments and add photos if necessary.

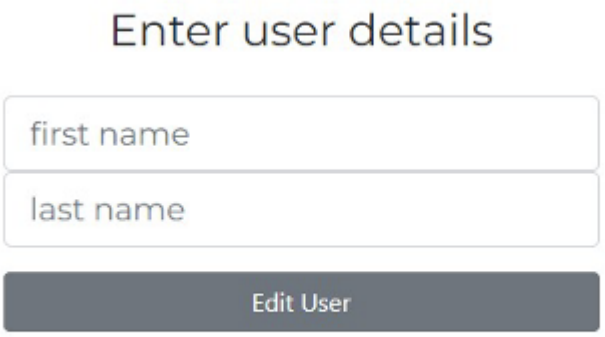
"Manager" user - a user who can provide alerts to users to perform maintenance, set up maintenance lists, define and create a knowledge library, view the information collected, and perform database questioning.

"Administrator" user - a user who can manage system usage credentials and user approval.

- Users can be created via add user page



- users can also be updated / deleted via update user page



1.3Usage scenarios (Use Cases)

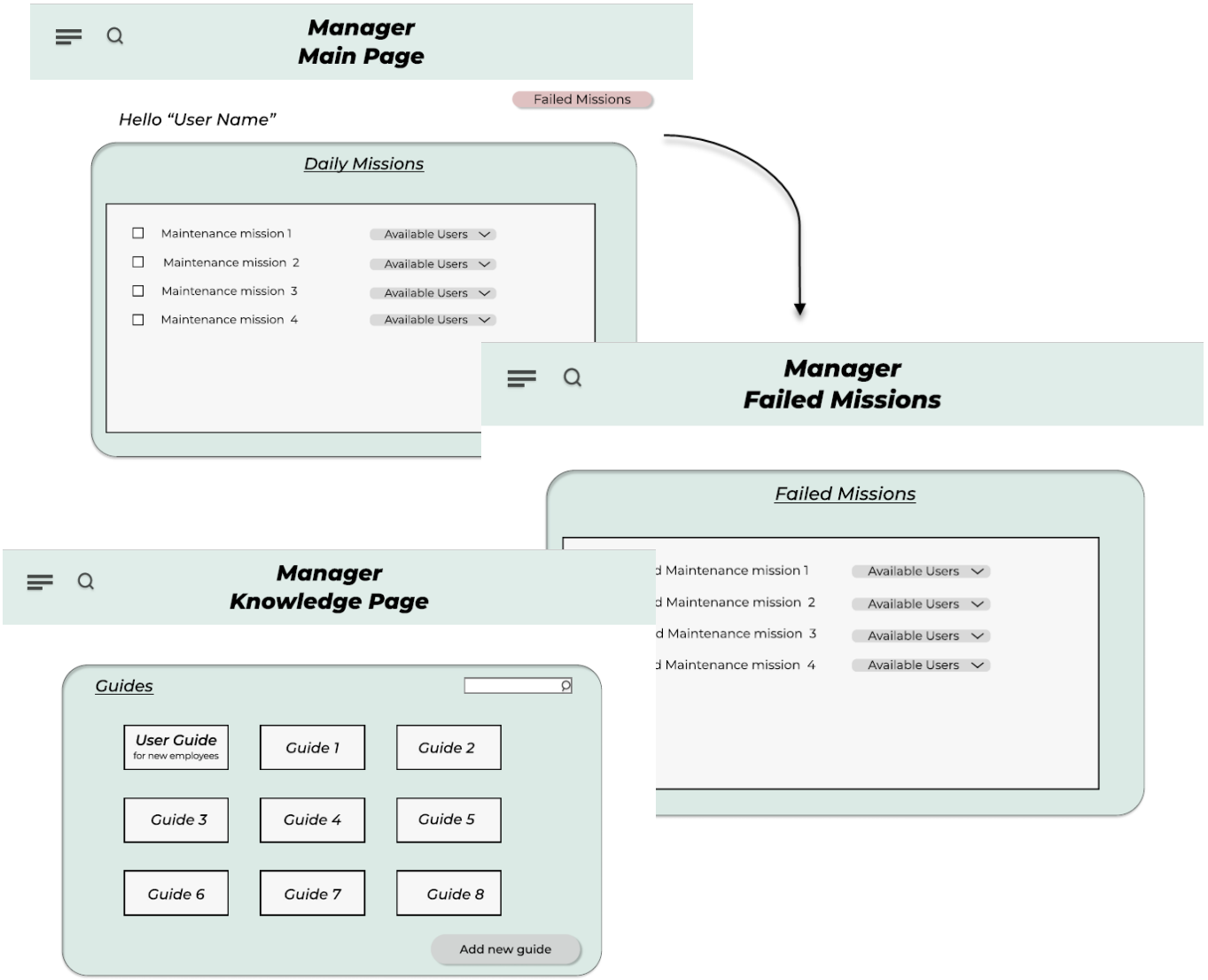
1.3.1 "Maintenance man" user Use Cases

- On the main page, the user will see all of his daily missions by the order of their arrival.
- When clicking on the "starting first mission" button, the system will navigate the user to the mission's location.
- When the system successfully identifies the location, the user will see:
  - All tasks to do in this location.
  - Maintenance history that includes – the last user that visits there, comments, photos from the location
- The user can check all of the subtasks.
- When a user is faced with a problem and can't complete the mission he will sign specific subtasks with X and will add a comment that will explain why he can't complete the task.
- When a user finishes his work at the location –
  - If he completes all subtasks – the mission will be completed & new maintenance date will add to the missions list.
  - If he failed to complete the mission – the mission will go back to the manager that will decide what to do about it.
- Users will repeat this process for the rest of the day or when all the missions are accomplished.
- User will have ‘Close day’ button - while clicking it all open missions in the same day will automatically moving to next day with higher frequency

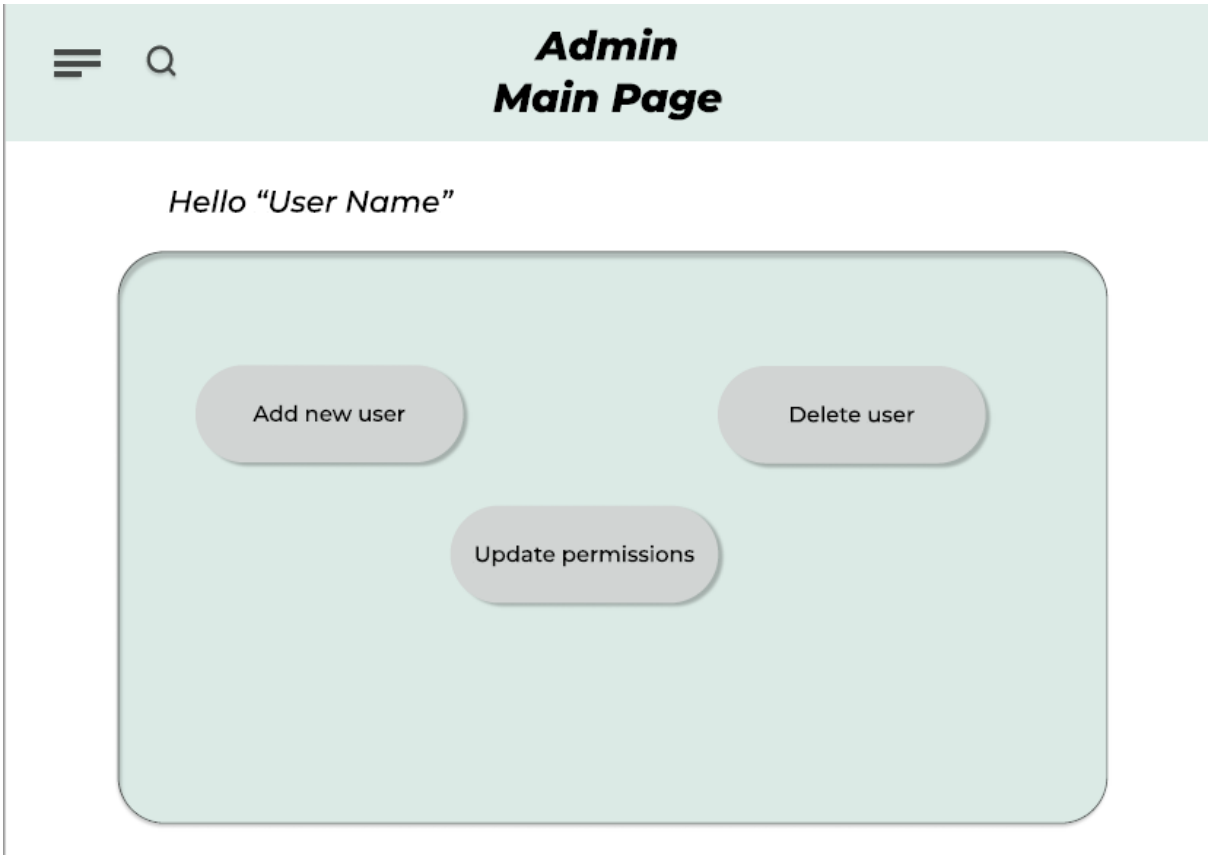
1.3.2 "Manager" user Use Cases

When the user opens the app at the beginning of the workday :

- On the main page see all open missions to his day
- Users will assign any mission to maintenance users base on their availability, location, and skills.
- On another page User will review all failed missions & figure out how to fix them – Re-assign to relevant users.
- User add & update Knowledge page that contains all maintenance missions that include all subtasks to each mission, Tools that needed to be complete, User guide to new employees, etc.

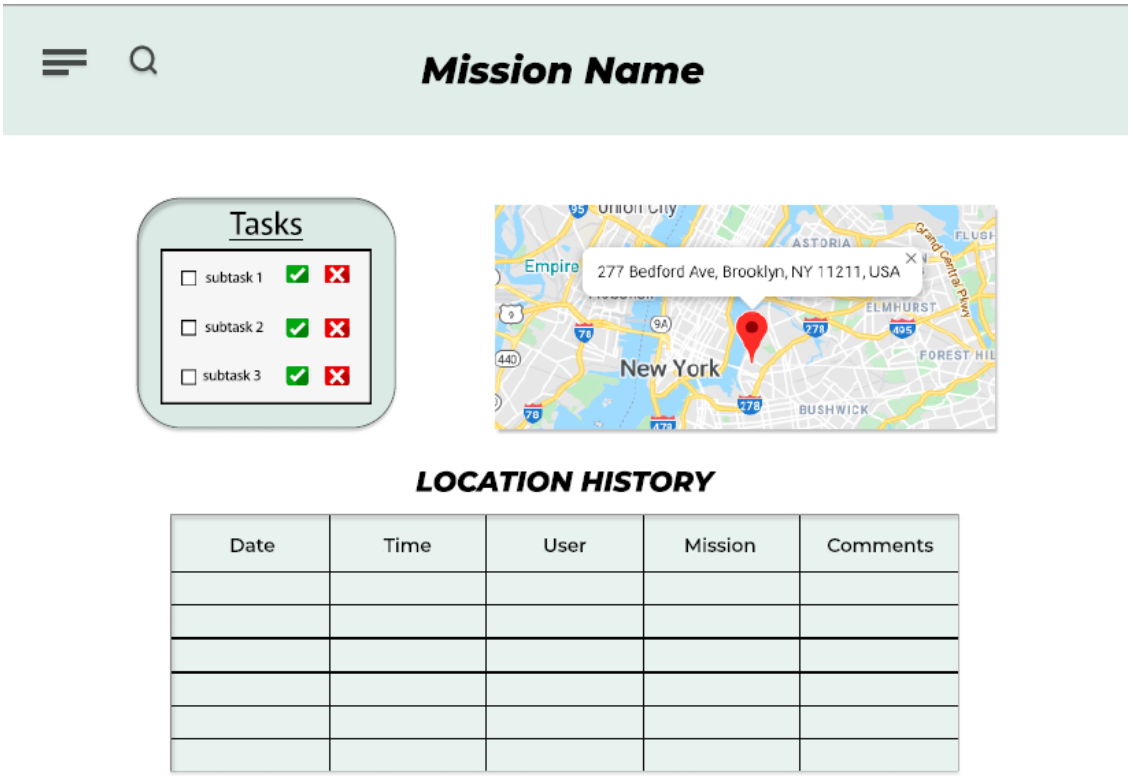


- That user will have the ability to add new users with specific permission (maintenance\manager\administrator) – for new employees.
- User can delete users for the system.
- Users can update permission to existing users, in case employees are promoted.



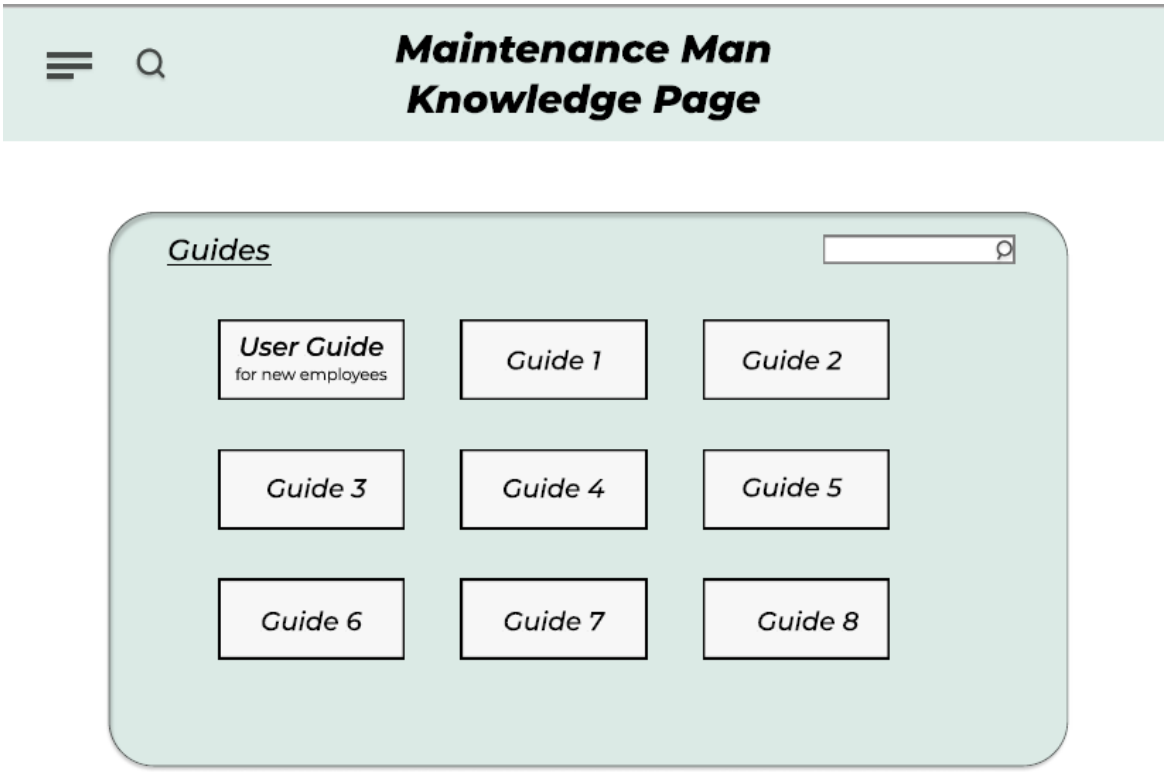
1.3.4 Maintanace mission menu

- **The manager user will be able to add Maintanace main missions and determinate the frequency and the sub-tasks that should be done as a checklist. Manager can assign any mission to any maintenance man.**
- **The maintenance man will be able to track his progress (in sub-tasks) by using checkboxes that will be located next to each sub-task.**
- **The maintenance man can open service tickets about any issues that he could face while executing the maintenance sub-tasks (this issue will automatically contain all relevant data - location ,time , description , etc.)**

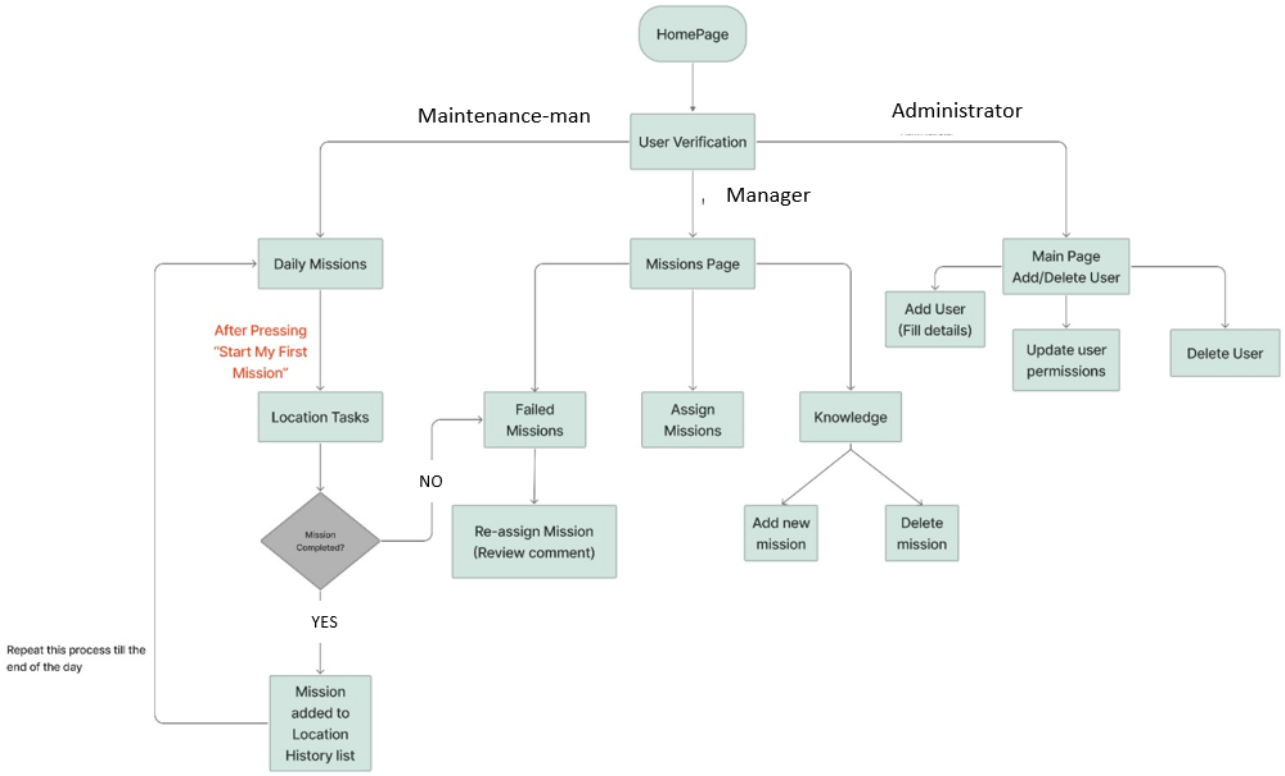


1.3.5 Maintanace knowlege page

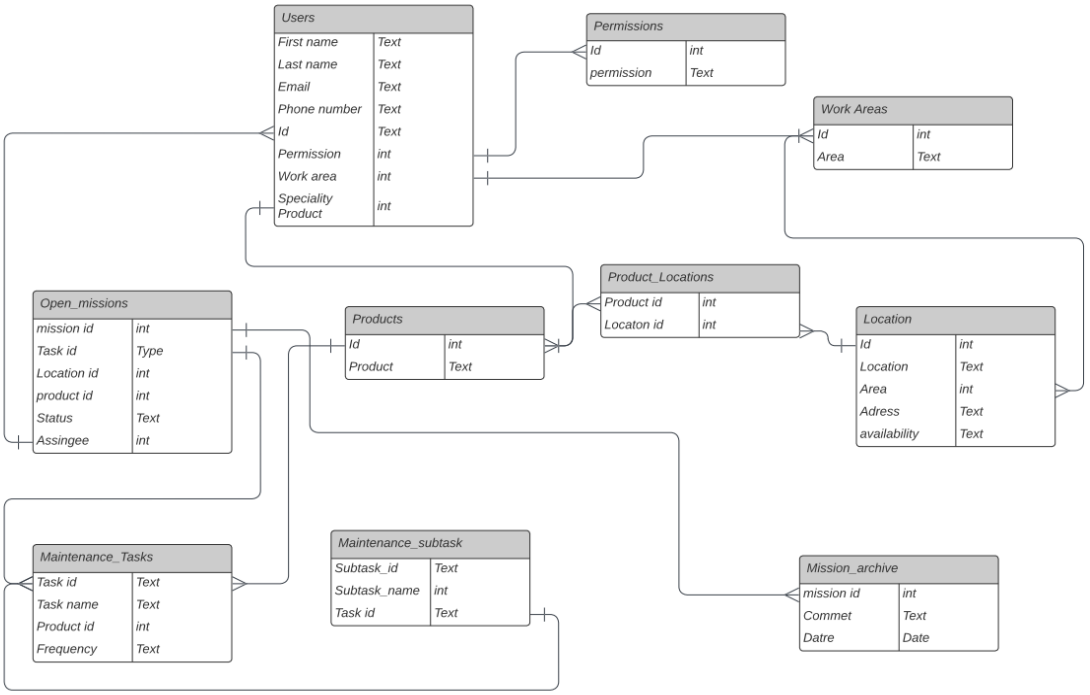
- **Maintanace man could write guide for any mission (can add tips , photos , videos that can help other workers to complete this mission easily)**
- **When maintenance man finish his guide the guide will move to the manager user that should approve it before all maintenance men will be able to see it.**
- **all employees can give feedbacks to any guide (the system will automatically re-order guides ,once a day , by avg rating / most recent guide)**


Maintenance Man
Knowledge Page
Guides
User Guide
for new employees
Guide 1
Guide 2
Guide 3
Guide 4
Guide 5
Guide 6
Guide 7
Guide 8

2. Work Flow


HomePage
User Verification
Maintenance-man
Administrator
Manager
Daily Missions
Missions Page
Main Page
Add/Delete User
Add User
(Fill details)
Update user
permissions
Delete User
After Pressing
"Start My First
Mission"
Location Tasks
Failed
Missions
Assign
Missions
Knowledge
NO
Mission
Completed?
Re-assign Mission
(Review comment)
Add new
mission
Delete
mission
YES
Repeat this process till the
end of the day
Mission
added to
Location
History list

3. DB tables


Users
First name Text
Last name Text
Email Text
Phone number Text
Id Text
Permission int
Work area int
Speciality Product int
Permissions
Id int
permission Text
Work Areas
Id int
Area Text
Product_Locations
Product id int
Locaton id int
Open_missions
mission id int
Task id Type
Location id int
product id int
Status Text
Assingee int
Products
Id int
Product Text
Location
Id int
Location Text
Area int
Adress Text
availability Text
Maintenance_Tasks
Task id Text
Task name Text
Product id int
Frequency Text
Maintenance_subtask
Subtask_id Text
Subtask_name int
Task id Text
Mission_archive
mission id int
Commet Text
Datre Date